Mental Health Needs and Treatment Access Disparities Among U.S. Adolescents

A Population Needs Assessment Using the 2024 National Survey of Children's Health

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1. Data and Cohort

This analysis uses the **2024 National Survey of Children's Health (NSCH) Topical file**, restricted to adolescents ages 12-17 (n = 21,960). Income is classified using the imputed Federal Poverty Level variable (`FPL_I1`), with **low income** defined as household income at or below 200% of the Federal Poverty Level (n = 5,674, 25.8% of the cohort).

> **Note on survey weighting:** NSCH uses a complex survey design. All estimates below are weighted using the provided final weight (`FWC`) and design variables (`STRATUM`, `HHID`) via the `survey` package, per Census Bureau/NSCH analytic guidance. Unweighted counts will not match published NSCH estimates — weighted estimates are what should be reported.

Weighted Prevalence of Current Mental Health Conditions, Adolescents 12-17

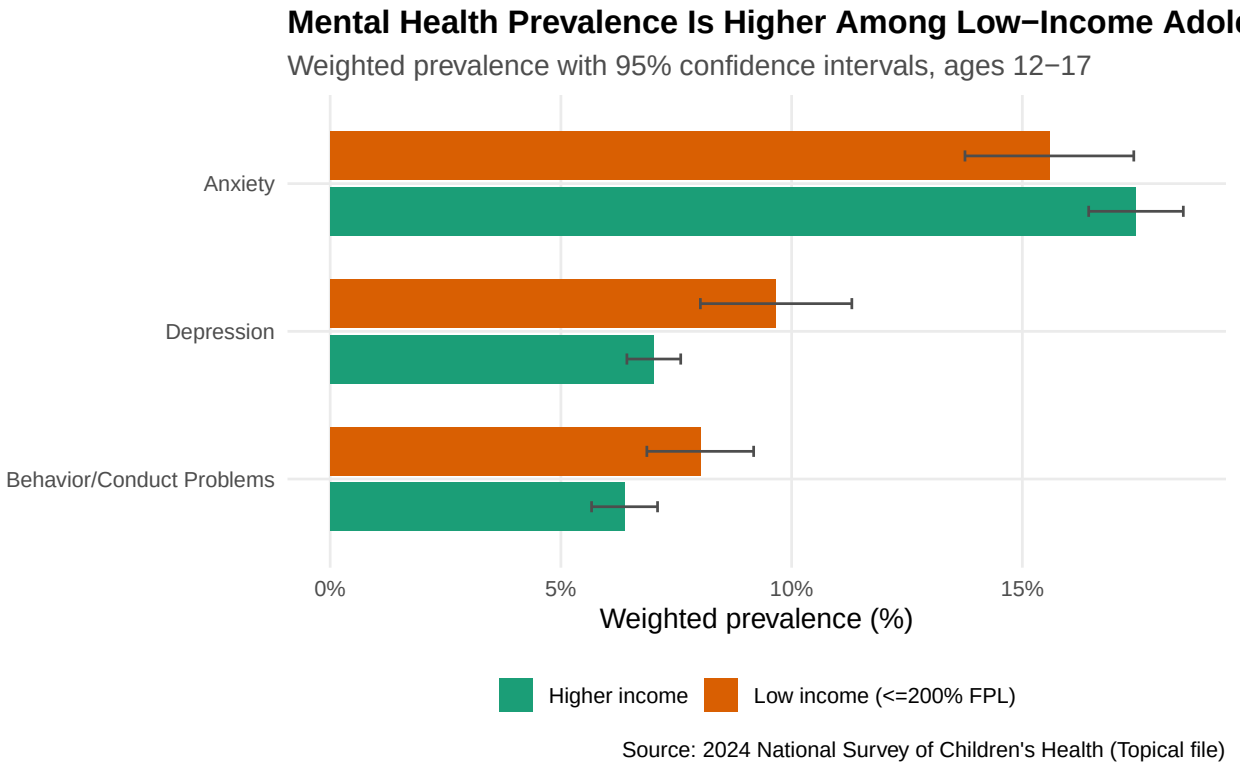
Group	Prevalence (%)
Depression	
Low income (200% FPL)	9.7% (±1.6)
Higher income	7.0% (±0.6)
Anxiety	
Low income (200% FPL)	15.6% (±1.8)
Higher income	17.5% (±1.0)
Behavior/Conduct Problems	
Low income (200% FPL)	8.0% (±1.2)
Higher income	6.4% (±0.7)

Source: 2024 NSCH Topical file. Estimates weighted using FWC; 95% CI shown in parentheses.

2. Mental Health Prevalence Overall and by Income

Table 1. Weighted Prevalence of Current Mental Health Conditions

Figure 1. Prevalence by Income Status



Unmet Mental Health Treatment Need, by Income Status

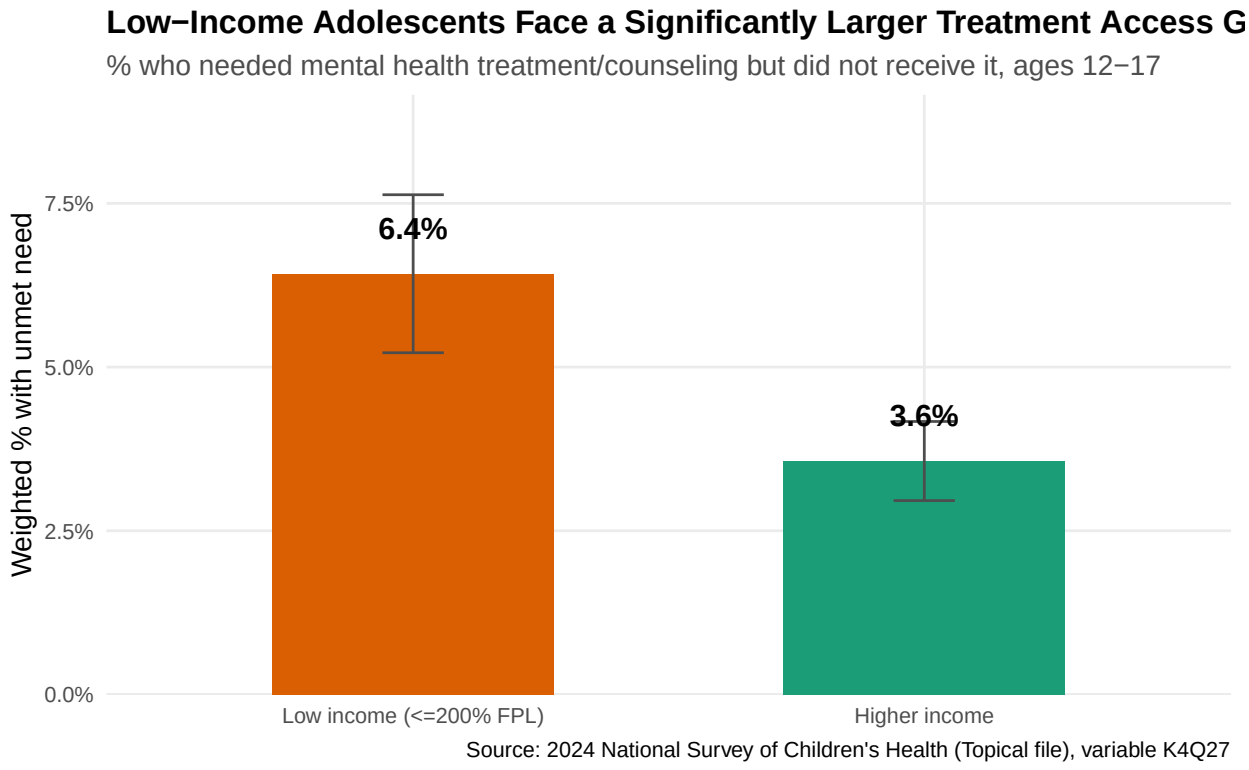
Group	Unmet Need (%)
Overall	4.6% (±0.6)
Low income (200% FPL)	6.4% (±1.2)
Higher income	3.6% (±0.6)

Source: 2024 NSCH Topical file, variable K4Q27. Estimates weighted; 95% CI shown in parentheses.

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3. Unmet Mental Health Treatment Need

Figure 2. The Treatment Access Gap

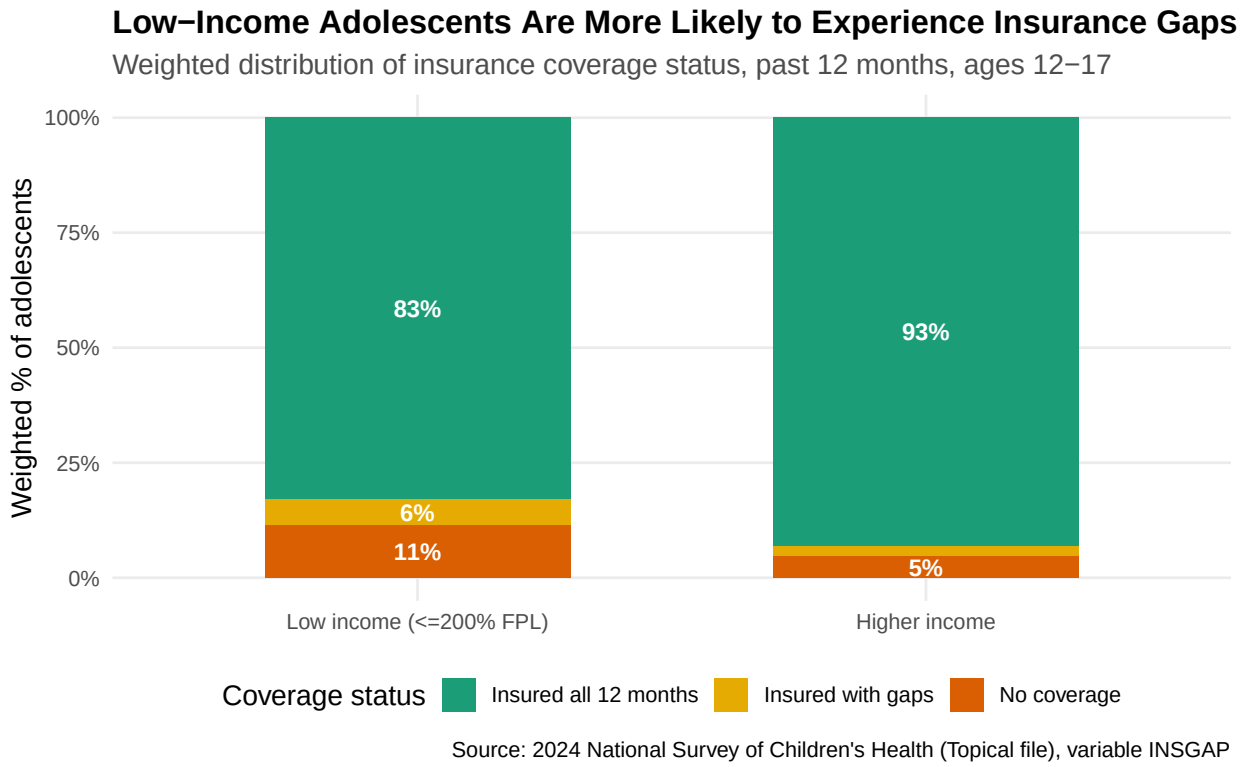


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4. Insurance Coverage Gaps

Unmet treatment need does not exist in isolation — it is closely tied to whether a child has continuous insurance coverage. This section examines coverage stability as a second, related access barrier.

Figure 3. Insurance Coverage Stability by Income Status



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5. Summary of Findings

- **16.8%** of adolescents currently have anxiety, with minimal difference by income status.
- **Depression** is notably more prevalent among low-income adolescents (**9.7%**) than higher-income adolescents (**7.0%**).
- **Behavior/conduct problems** show a similar income-linked gap (**8.0%** vs. **6.4%**).
- Low-income adolescents are roughly **1.8x** as likely to report an unmet mental health treatment need (**6.4%**) compared to higher-income adolescents (**3.6%**).
- Low-income adolescents are also more likely to experience **coverage gaps or a full lack of insurance**, compounding the treatment access barrier documented above.

6. Design Implications for the Digital Intervention

1. **Prioritize depression and behavioral/conduct support** as the intervention's primary content focus — these show the clearest income-linked prevalence disparities, unlike anxiety, which is elevated but roughly flat across income groups.
2. **Frame the tool's value proposition around closing the treatment access gap**, not only symptom reduction — the data supports "harder to access care" as the specific, measurable problem for the target population, not just "higher symptom burden."
3. **Anxiety content can still be included** given its high overall prevalence (~20%), but should not anchor the underserved-population framing, since it is not disproportionately concentrated among low-income adolescents in this data.

4. **Design for insurance-gap resilience** — since low-income adolescents are more likely to have unstable or absent coverage, the intervention should not assume a stable clinical referral pathway is always available, and should function as a standalone support option, not solely a bridge to insurance-dependent care.

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7. Discussion: Connecting These Findings to the RDoC-Guided AI Framework

The findings in this brief serve a dual purpose within the broader project: they establish the population-level need justifying this intervention, and they illustrate, in a very concrete way, the exact limitation of diagnosis-based data that our companion prospect paper — *"The Research Domain Criteria (RDoC) as a Roadmap for Precision AI in Mental Health"* — argues AI-guided mental health tools should move beyond.

What this data can tell us, and what it cannot. NSCH measures depression, anxiety, and behavior/conduct problems as parent-reported, clinician-diagnosed categories — the same DSM/ICD-style diagnostic labels our paper's Introduction (Section 3.2) identifies as the "ground truth problem" for AI development. This needs assessment shows *that* low-income adolescents experience elevated depression and behavior/conduct problems and a substantially larger treatment access gap. What it cannot show is *why*, at a mechanistic level: whether the elevated depression prevalence in this subgroup reflects, for example, blunted reward responsiveness (RDoC Positive Valence Systems), heightened and sustained threat processing linked to economic instability (RDoC Negative Valence Systems), or disrupted sleep and arousal regulation associated with household stress (RDoC Arousal/Regulatory Systems). A diagnostic label of "depression" or "behavior/conduct problems" can be the downstream expression of any of these distinct underlying processes, or some combination of them, and a population survey built on diagnostic categories cannot distinguish between them.

This is precisely the gap Section V of the paper argues RDoC-guided AI is positioned to close. Where this needs assessment identifies *which population* and *which broad diagnostic categories* to prioritize, an RDoC-informed digital intervention would go a level deeper for the individual user, using passively collected data, conversational patterns, or engagement metrics to infer *which specific mechanistic construct* is driving that user's presentation, and adapt content accordingly. Concretely, this data motivates two direct connections to our paper's sections:

- **Section 4.2.1 (Chatbots and Virtual Assistants):** The treatment access gap finding (low-income adolescents facing roughly double the unmet need of higher-income peers) is the strongest empirical justification in this brief for a conversational AI delivery model specifically. A chatbot-delivered intervention does not depend on insurance coverage, appointment availability, or transportation, all of which this data suggests are more constrained for the target population (see Figure 3). This reframes the chatbot literature reviewed in 4.2.1 from "a nice-to-have alternative modality" to "a structurally necessary one for this specific population."
- **Section 5.1 and 5.2 (Negative and Positive Valence Systems):** Since this population-level data cannot distinguish threat-driven versus reward-driven pathways to depression, it directly motivates building the content module (Deliverable 1's applied output) to incorporate RDoC-informed screening or adaptive branching, rather than a one-size-fits-all "depression module," so that the tool itself can begin to capture the mechanistic granularity this survey data cannot.

In short, this needs assessment answers the epidemiological question the paper's Introduction poses (is there a real, measurable disparity worth building for), while simultaneously reproducing, in miniature, the exact diagnostic-category limitation the paper's central thesis is built to address, which strengthens rather than weakens the case for the RDoC-guided approach described in Sections V and VI.

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8. Methods Note

Data are from the 2024 NSCH Topical file, restricted to adolescents ages 12–17. All prevalence estimates are weighted using the survey's final weight (`FWC`) with stratification (`STRATUM`) and household clustering (`HHID`), consistent with NSCH's recommended complex survey design approach. Income is classified using the first imputed Federal Poverty Level variable (`FPL_I1`); NSCH provides six multiply-imputed FPL variables (`FPL_I1`–`FPL_I6`) to address item non-response in income reporting, and a full multiple-imputation combination (Rubin's rules) across all six is recommended for a peer-reviewed manuscript, though a single implicate is generally adequate for an internal practicum data brief. Insurance coverage status is derived from `INSGAP`, which categorizes coverage over the past 12 months as continuous, gapped, or absent.